

WS N°1: Diffusion

1. A sheet of steel 1.5 mm thick has nitrogen atmospheres on both sides at 1200°C and is permitted to achieve a steady-state diffusion condition. The diffusion coefficient for nitrogen in steel at this temperature is $6 \times 10^{-11} \text{ m}^2/\text{s}$, and the diffusion flux is found to be $1.2 \times 10^{-7} \text{ kg/m}^2 \cdot \text{s}$. Also, it is known that the concentration of nitrogen in the steel at the high-pressure surface is 4 kg/m^3 . How far into the sheet from this high-pressure side will the concentration be 2 kg/m^3 ? Assume a linear concentration profile.
2. The purification of hydrogen gas by diffusion through a palladium sheet is performed. Compute the number of kilograms of hydrogen that pass per hour through a 5 mm thick sheet of palladium having an area of 0.2 m^2 at 500°C. Assume a diffusion coefficient of $1 \times 10^{-8} \text{ m}^2/\text{s}$, that the concentrations at the high- and low-pressure sides of the plate are 2.4 and 0.6 kg of hydrogen per cubic meter of palladium, and that steady-state conditions have been attained.
3. A sheet of BCC iron 1 mm thick was exposed to a carburizing gas atmosphere on one side and a decarburizing atmosphere on the other side at 725°C. After having reached steady state, the iron was quickly cooled to room temperature. The carbon concentrations at the two surfaces of the sheet were determined to be 0.012 and 0.0075 wt%. Compute the diffusion coefficient if the diffusion flux is $1.4 \times 10^{-8} \text{ kg/m}^2 \cdot \text{s}$. The densities of carbon and iron are respectively 2.25 g/cm^3 and 7.87 g/cm^3 .
4. Determine the carburizing time necessary to achieve a carbon concentration of 0.45 wt% at a position 2 mm into an iron–carbon alloy that initially contains 0.2 wt% C. The surface concentration is to be maintained at 1.3 wt% C, and the treatment is to be conducted at 1000°C. Use the diffusion data for $\gamma\text{-Fe}$ from the table.
5. For a steel alloy it has been determined that a carburizing heat treatment of 10 hrs duration will raise the carbon concentration to 0.45 wt% at a point 2.5 mm from the surface. Estimate the time necessary to achieve the same concentration at a 5 mm position for an identical steel and at the same carburizing temperature.
6. Calculate the values of the diffusion coefficients for the interdiffusion of carbon in both α -iron (BCC) and γ -iron (FCC) at 900°C. Which is greater? Justify your answer.
7. At which temperature will the diffusion coefficient for the diffusion of copper in nickel have a value of $6.5 \times 10^{-17} \text{ m}^2/\text{s}$.

8. The steady-state diffusion flux through a metal plate is $5.4 \times 10^{-10} \text{ kg/m}^2 \cdot \text{s}$ at a temperature of 727°C and when the concentration gradient is -350 kg/m^4 . Calculate the diffusion flux at 1027°C for the same concentration gradient and assuming an activation energy for diffusion of 125 KJ/mol .